



Experimental and numerical investigation of clearance gap effects on laminar mixed convection heat transfer from fin array in a horizontal channel-A conjugate analysis

M. Dogan^a, M. Sivrioglu^{b,*}

^a Department of Mechanical Engineering, Bozok University, 66100 Yozgat, Turkey

^b Department of Mechanical Engineering, Gazi University, 06570 Maltepe, Ankara, Turkey

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ABSTRACT

The influence of the clearance gap between fin tips and the top surface of channel on laminar mixed convection heat transfer from longitudinal fins has been experimentally and numerically investigated. A conjugate analysis has been carried out in which the conservation of mass, momentum and energy equations for the fluid in the two fin enclosure are solved together with the heat conduction equation in the fin and the base plate. Average heat transfer coefficients were obtained for modified Grashof numbers $5 \times 10^7 < Gr^* < 1 \times 10^9$, and dimensionless fin spacing was varied from $S/H = 0.04$ to $S/H = 0.18$ and clearance parameter was varied from $c/H = 0.20$ to $c/H = 0.75$, to determine the effect of the clearance gap. The velocity of fluid entering the channel was varied between $0.02 \leq w_{in} \leq 0.23$ m/s so that experiments were conducted for Reynolds numbers $250 < Re < 2300$. The results obtained from this study show that the heat transfer coefficient increases with the decrease in the clearance parameter. As a result of experimental study, an empirical equation has been derived to correlate the average Nusselt number as a function of Reynolds number, modified Grashof number and fin spacing, fin height, tip clearance and fin length. This equation is also in agreement with the numerical results.

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1. Introduction

Heat transfer is very important for the proper design and reliable operation of systems. The reliability and performance of a system are highly dependent on the system operating temperature, especially in electronic equipment and similar industrial applications. Fins are often used to enhance the rate of heat transfer from heated surfaces to the environment. They can be placed on plane surfaces, tubes, or other geometries. The use of extended surfaces such as fins provides a reliable, cheap and widely used method of cooling for dissipating heat. When an array of fins is used to enhance heat transfer, the optimum geometry of fins (corresponding to a maximum rate of heat transfer) should be used, provided it is compatible with available space and financial limitations. Advantages of printed circuits boards have yielded increasing power dissipation from surfaces in a channel. Providing adequate cooling of printed circuits boards has recently motivated experiments on the use of longitudinal fins to enhance heat transfer in rectangular channels. The heat transfer to the fluid flowing through a channel by the heat dissipating surfaces

occurs mainly due to the mechanisms of heat transfer by forced convection, natural convection and by radiative heat transfer.

An extensive review and discussion of work done on the convective heat transfer in electronic cooling were presented by Incropera [1], summarizing various convection cooling options. A great number of experimental and analytical works has been carried out on this problem since Elenbaas [2], who first introduced the problem of natural convection between parallel plates. Starner and McManus [3], who measured the average heat transfer coefficient not only in horizontal but also in 45° and vertical base positions, have performed the first work on horizontal rectangular fin arrays. They showed that incorrect application of fins to a surface actually might reduce the total heat transfer to a value below that of the base alone. Welling and Wooldridge [4] conducted a similar experimental study on vertical fins to determine maximum heat transfer and tried to find optimum values for the ratio of fin height to fin spacing. Harahap and McManus [5] observed the flow patterns in two series of horizontal fin arrays. Jones and Smith [6] investigated the effects of fin height and fin spacing on the heat transfer coefficient. They concluded that fin spacing is the main geometrical parameter, and it should be chosen as characteristic length. Numerous experimental studies exist in

* Corresponding author. Tel.: +90 312 5823448; fax: +90 312 2308434.
E-mail address: mecits@gazi.edu.tr (M. Sivrioglu).

Nomenclature

A_b	copper plate surface area, m ²
A_c	channel cross sectional area, m ²
A_R	radiation heat transfer surface area, m ²
C	clearance parameter, $C = c/H$
c	clearance gap, m
D_h	channel hydraulic diameter, $D_h = A_c/P$, m
F	gray body shape factor
g	gravitational acceleration, m/s ²
Gr^*	modified Grashof number $Gr^* = \frac{g\beta q_{con} D_h^4}{k_{air} \nu_{air}^2}$
h_{av}	average convective heat transfer coefficient, W/m ² K
H	channel height, m
H_f	fin height, m
I	electric current
k	thermal conductivity, W/mK
L	length, m
Nu_{av}	dimensionless average Nusselt number, $Nu_{av} = h_{av} D_h / k_{air}$
P	perimeter, m
q_{con}	convection heat flux, W/m ²
$Q_{convection}$	convection heat transfer rate, W
$Q_{conduction}$	conduction heat transfer rate, W
$Q_{radiation}$	radiation heat transfer rate, W
Q_{total}	total power dissipation, W

Ra	Rayleigh number, $Ra = (g\beta\Delta TH^3)/(\alpha_{air}\nu_{air})$
Re	Reynolds number, $Re = w_{in} D_h / \nu_{air}$
T	temperature, °C
T_b	bulk mean temperature, °C
T_{in}	inlet air temperature, °C
T_w	average copper plate surface temperature, °C
S	fin spacing, m
w_{in}	mean longitudinal velocity, m/s
V	voltage, V
W	width, m

Greek symbols

α	thermal diffusivity, m ² /s
β	thermal expansion coefficient, 1/K
ε	heater surface emissivity
μ	dynamic viscosity, kg/ms
ν	kinematic viscosity, m ² /s
σ	Stefan–Boltzmann constant, W/m ² K ⁴

Subscripts

air	air
f	fin
in	inlet
w	wall

the literature, that were carried out in recent years on natural convection heat transfer from rectangular fin arrays placed on either horizontal or vertical plates [7–11]. In these studies effects of fin height, fin spacing, optimum fin spacing and the difference between the base surface temperature and ambient air temperature were investigated. As results of these studies some correlation equations were proposed as a function of fin height and spacing to obtain the maximum heat transfer in the case of natural convection with fin arrays for a certain temperature difference between the fin base and ambient air. A survey of literature reveals some theoretical studies on horizontal fin arrays [12–18]. Radiation heat transfer plays an important role in heat transfer from fin arrays. Some studies also exist in literature, which take into account the effects of radiation on convective heat transfer from fin arrays by Ellison [19].

An analytical study was made to investigate the effects of buoyancy on laminar mixed convection in a shrouded fin array by Acharya and Patankar [20]. They concluded that the buoyancy forces significantly affect the heat transfer characteristics of laminar mixed convection in a shrouded fin array, buoyancy forces induce secondary flows and the increase in effects of natural convection, causes the buoyancy driven secondary flow to develop strongly and therefore, causes an increase in the rate of heat transfer. Laminar mixed convection in tubes with longitudinal internal fins was investigated experimentally by Rustum and Soliman [21]. Results of experiments showed that natural convection has a strong influence on heat transfer in finned tubes and the enhancement in Nusselt number due to natural convection starts at high values of Rayleigh number. The effects of fin geometries and the fin tip to shroud clearance on the heat transfer from the longitudinal rectangular fin array in a horizontal channel were investigated experimentally [22,23]. Kadle and Sparrow [24] investigated fully developed heat transfer from a shrouded longitudinal fin array without tip clearance to a turbulent flow experimentally and numerically. They found that the local heat transfer coefficient varied along the fin and base plate surface. The lowest value of the heat transfer coefficients was found at the intersection

of fin base and fin-shroud. Sparrow and Kadle [25], who considered the influence of the tip to shroud clearance for turbulent flow and found that heat transfer coefficient decreases with increasing clearance, because increasing clearance, delayed the achievement of fully developed conditions. The heat transfer from a shrouded longitudinal fin array with, and without tip clearance was investigated by Sparrow, Baliga and Patankar [26]. They observed that the heat transfer takes a minimum next to the base and increases toward the fin tips. The maximum heat transfer occurs either at the tip or the intermediate place between the base and the tip.

From the above review of literature one can see that only little information can be found in literature for the influence of clearance gap on heat transfer from longitudinal fins inside a horizontal channel. A few mixed convection studies were carried out in channels at constant base plate temperatures. To achieve maximum heat transfer for a constant heat flux condition, the selection of fin spacing must be optimized. The present study aims to numerically and experimentally investigate heat transfer from rectangular fin arrays, which are mounted on the bottom wall of a horizontal channel. The effects of the fin height or the clearance parameter, and fin spacing on laminar mixed convection heat transfer are investigated numerically. The experimental results are used for the verification of numerical model. During the experiments constant heat flux thermal boundary condition was realized, and air was used as the working fluid.

2. Mathematical formulation and numerical model

In this section, the description of the problem used in the study is given, the mathematical formulation and boundary conditions are shortly described.

2.1. Problem description

A schematic drawing of the fin array under investigation and the computational domain used for numerical simulations are shown in Fig. 1.

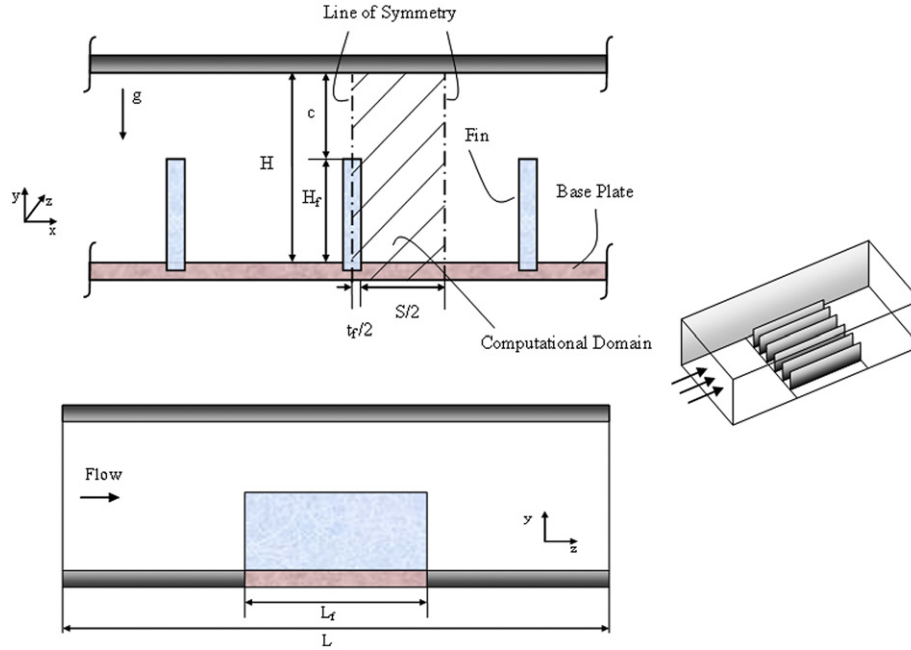


Fig. 1. Schematic drawing of fin array and computational domain.

2.2. Governing equations

The temperature and velocity fields in the region between two fins are governed by conservation of mass, momentum and energy equations for the fluid together with the three-dimensional heat conduction equation for the fin and base plate, which are given below. Fluid properties were taken constants except the buoyancy term. The base plate and fin properties were assumed to remain constant. Thus the equations of 3-D, steady, laminar and incompressible flow considered in this study are as follows.

Conservation of mass:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (1)$$

Conservation of momentum:

$$\left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{1}{\rho} \frac{\partial P}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \quad (2)$$

$$\left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = -\frac{1}{\rho} \frac{\partial P}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) + \beta g (T - T_0) \quad (3)$$

$$\left(u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = -\frac{1}{\rho} \frac{\partial P}{\partial z} + \nu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) \quad (4)$$

Conservation of energy:

$$\left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) = a \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) \quad (5)$$

Heat conduction equation for base plate and fin:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0 \quad (6)$$

During calculations, the ambient temperature T_0 was selected as $T_0 = T_{in}$. The fluid properties of air were determined at this temperature.

2.3. Solution algorithm

The governing equations are solved by the widely used CFD package FLUENT 6.3.26, which employs a finite volume method. The velocities and the pressure are calculated using the semi implicit pressure linked equation solver (SIMPLE) algorithm. The convective terms are discretized using a second-order upwind scheme.

2.4. Boundary conditions

Because of symmetry conditions and only one fin channel is investigated the resulting computational domain and the boundaries of the system are presented as in Fig. 2. The conduction equations are solved in the walls consisting of the fin (KJMN

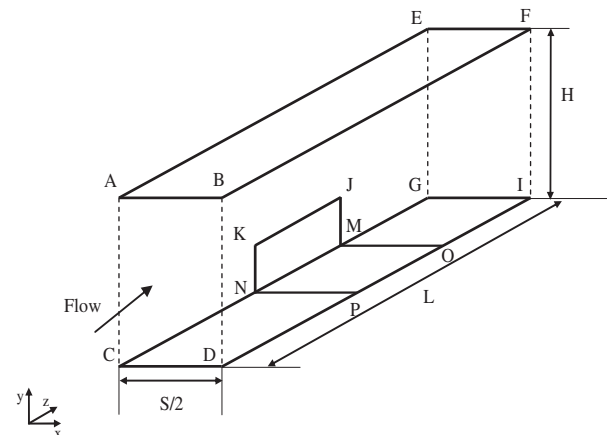


Fig. 2. A schematic drawing of the computational domain showing locations of the applied boundary conditions.

surface) and the base plate (NMOP surface). The other surfaces are the top wall (ABFE surface), the unheated entrance region (CDPN surface) and exit region (OMGI surface). No slip condition was applied along all solid walls and the flow velocities on these walls were set to zero. The constant heat flux thermal boundary condition was applied at the base plate with an adiabatic top surface. Symmetry boundary conditions were applied at the surfaces of BDIF and ACEG. The inlet and outlet boundary conditions were applied at the surfaces of ABDC and EFIG, respectively. The applied boundary conditions are as given below.

Uniform velocity and temperature at the inlet:

$$u = v = 0, w = w_{in}, T = T_{in} \quad (7)$$

at the outlet:

$$\frac{\partial u}{\partial z} = 0, \frac{\partial v}{\partial z} = 0, \frac{\partial w}{\partial z} = 0, \frac{\partial T}{\partial z} = 0, \quad (8)$$

At the symmetry axis:

$$\frac{\partial u}{\partial x} = 0, \frac{\partial v}{\partial x} = 0, \frac{\partial w}{\partial x} = 0, \frac{\partial T}{\partial x} = 0, \quad (9)$$

The heated base plate (NMOP surface):

$$k_s \frac{\partial T}{\partial y} = q_0 \quad (10)$$

The points of solid–fluid contact must maintain continuity of both heat flux and temperature so that

$$k_s \frac{\partial T}{\partial n} = k_f \frac{\partial T}{\partial n} \quad (11)$$

and

$$T_{solid} = T_{fluid} \quad (12)$$

In above equations, k_s and k_f are thermal conductivity of the solid wall surface and the air near the surface, respectively. q_0 represents the heat flux applied to the base plate.

2.5. Convergence, grid independency and comparison with experimental results

The convergence criterion was based on the residual value of the calculated variables, i.e., mass, velocity components and energy. The specified value was 10^{-4} for continuity and 10^{-7} for momentum and energy. The solutions obtained by FLUENT were tested for grid independency by solving the problem with the number of nodes varying from $5 \times 20 \times 200$ to $10 \times 70 \times 550$ and numerical results were obtained as shown in Table 1. It is seen that increasing the number of nodes, the average temperature of base plate changes up to $10 \times 60 \times 500$, and then it remains almost constant with the increase in the number of nodes. Thus final

simulations were performed with the number of nodes of $10 \times 60 \times 500$ in the x–y–z coordinate directions, respectively. The solution is independent of the number of nodes used in the fin material. Changing the number of nodes within the fin from 2 to 6 does not cause an important change in the average temperature of base plate. Hence, two nodes in the x direction of the fin were used for the fin thickness of $t/2 = 0.5$ mm.

Grids were refined close to the fin and base plate surface. The detailed view of the computational domain and grid distribution is shown in Fig. 3.

To validate the solution, results of experimental studies conducted by Maughan and Incropera [27] and Dogan [28] were compared with the present numerical results. The comparison of experimental and numerical results in terms of Nu number is shown in Fig. 4 and Fig. 5.

In Fig. 4, the Rayleigh number (Ra) is defined by:

$$Ra = \left(g \beta \Delta T H^3 \right) / (\alpha_{air} \nu_{air}) \quad (13)$$

The modified Grashof number (Gr^*) is defined by the equation (21). The values of Nusselt number for the present numerical study and for other experimental studies, however, were calculated by using the definition given in the equation (14).

As can be seen from Figs. 4 and 5, there is a good agreement between the numerical and experimental results. The maximum difference between experimental and numerical values doesn't exceed 12%. This can be considered as a sufficient verification of the computational domain and numerical procedures applied in the present study. The differences between the experimental and the numerical results, however, can be mainly due to the uncertainty in the Nusselt number which is stated in the last paragraph of "3.2. Processing of the experimental data".

3. Experimental setup and data reduction

Detailed information on the experimental apparatus, devices used and procedures followed are given by Dogan [28]. A summary of information is presented below.

3.1. Experimental setup

A schematic drawing of the experimental setup is shown in Fig. 6. The experimental setup consists mainly of a filter, a nozzle with flow straighteners, the test section, a damping chamber, a diffuser, a flow control valve and an AC fan. A data acquisition system with a PC was used for the measurement and storage of temperatures. A variac was used to adjust the electric current supplied to the heater plate. The test section is a rectangular duct heated at the bottom with a cross section of 300 mm in width and 100 mm in height. The top, sides and bottom, entry and exit regions of the duct are constructed of five mm thick plexi-glass. The unheated entrance region of the duct is 600 mm, the heated test section is 600 mm, and unheated exit region is 300 mm. At the channel entrance, a nozzle with flow straighteners has been used to suppress turbulence and achieve steady, laminar flow conditions with a uniform velocity distribution. The flow straightener was made of five mm diameter and 50 mm long plastic hoses. In addition, a filter was used to filter incoming air. The nozzle is made of 0.5 mm thick aluminum sheet. The test section consists of a $5 \times 300 \times 600$ mm copper plate, aluminum longitudinal fins of one mm in thickness and a heater plate and thermocouples. Aluminum fins were attached to the copper plate by machining longitudinal grooves one mm wide and 2.5 mm deep. Initially, the grooves were filled with a thermally-conductive paste. Then the fin edges were coated with conducting paste, and the fins were

Table 1
Variation of average base plate temperature with number of nodes.

Number of simulations	Node number	Average base plate temperature
1	$5 \times 20 \times 200$	35.18
2	$6 \times 30 \times 250$	35.24
3	$6 \times 40 \times 325$	34.69
4	$7 \times 50 \times 400$	35.66
5	$9 \times 60 \times 450$	35.87
6	$10 \times 60 \times 500$	36.12
7	$10 \times 65 \times 525$	36.27
8	$10 \times 70 \times 550$	36.31

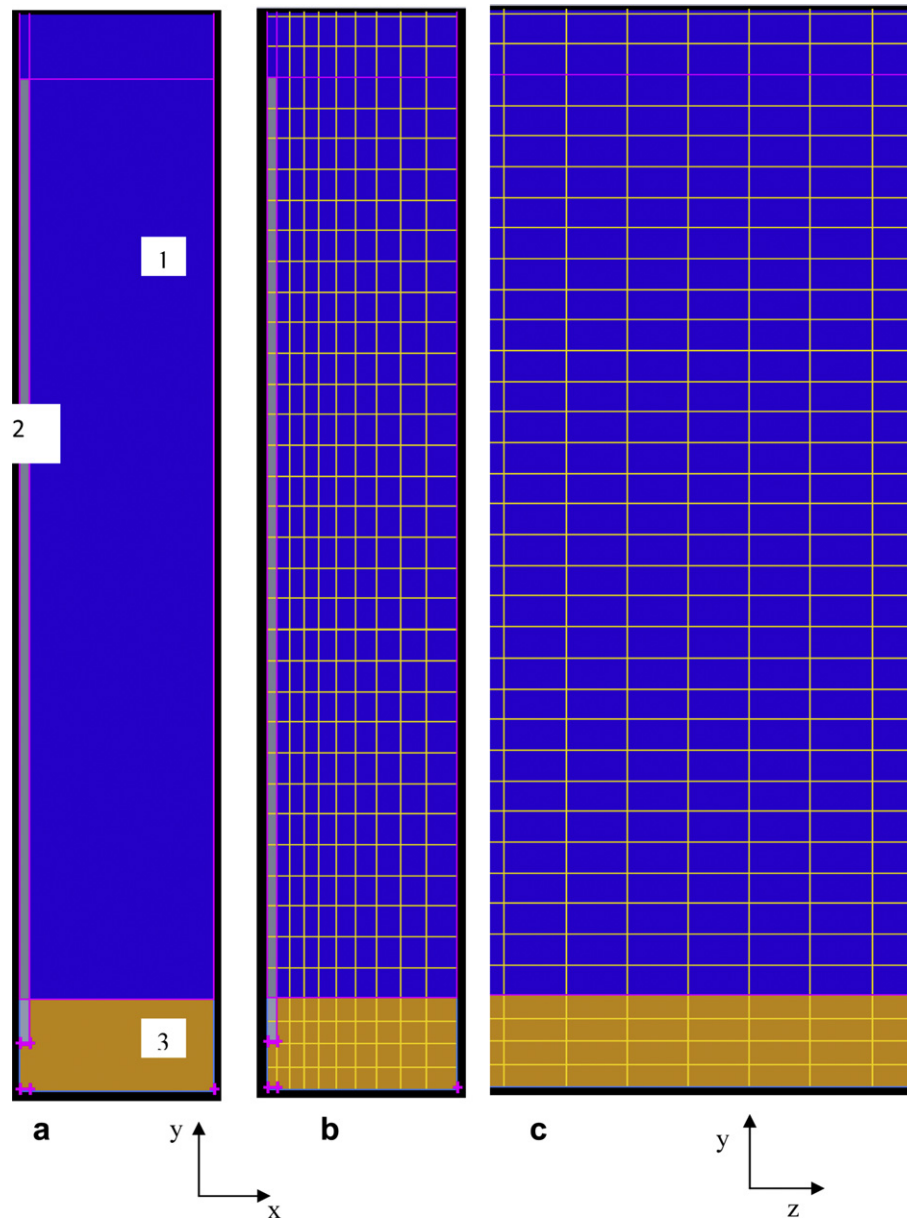


Fig. 3. Detail views of (a) Computational domain (1-fluid, 2-fin and 3-bottom plate) and grid distribution on (b) x–y plane (c) y–z plane.

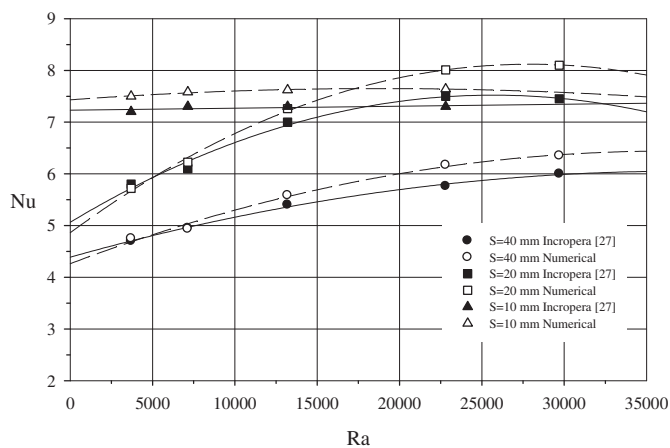


Fig. 4. Validation of the computational approach: the present numerical results compared with the experimental results obtained by Maughan and Incropera [27].

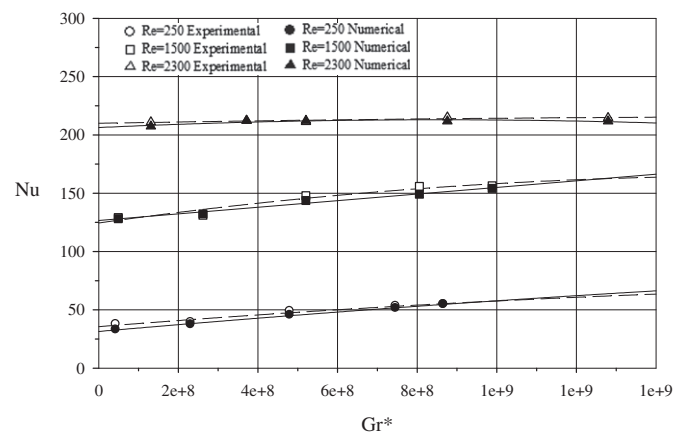


Fig. 5. Comparison of numerical and experimental results Dogan [28] for different Reynolds numbers for $C = 0.25$ and $S = 8$ mm.

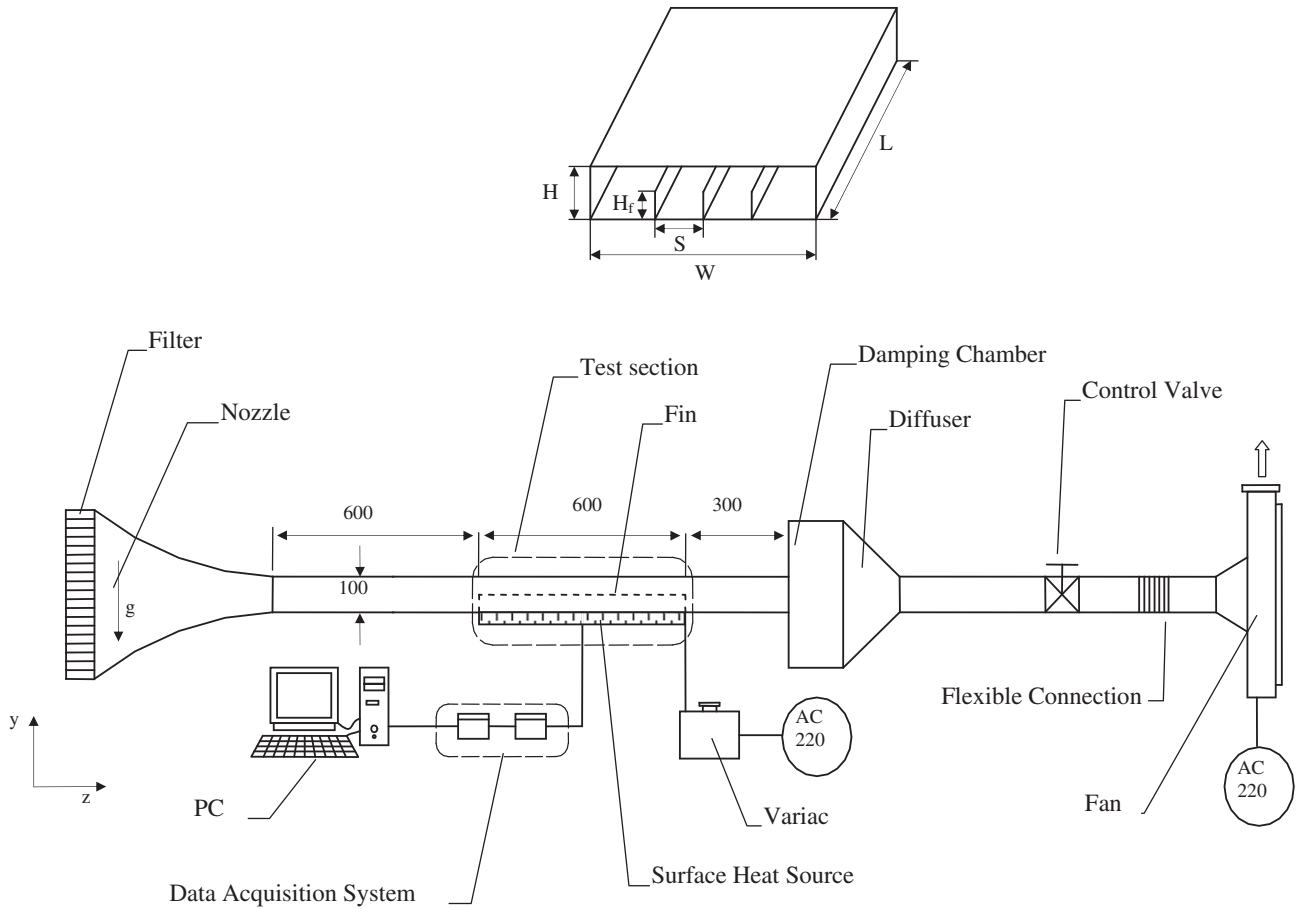


Fig. 6. Schematic diagram of the experimental set up, (measures are in mm).

pressed fitted into the length of the slots. The copper plate surfaces were polished to minimize radiation losses. The surface emissivities were measured to be 0.05 and 0.04 for the polished copper and aluminum surfaces, respectively. All surfaces were checked for flatness and were carefully cleaned and polished. A sheet heater plate was placed under the copper plate, with a size equal to the size of copper plate. Electric current was provided to the heater plate via a variac, providing a heat flux boundary condition specified for a decided experimental case. The cross section view of the rectangular duct is schematically shown in Fig. 7. As can be seen from this figure, the test section of the duct was insulated with 15 mm Glass wool ($k = 0.048$ W/m K) and 50-mm Styrofoam ($k = 0.028$ W/m K) and was mounted on a rigid supporting frame. The test section is isolated from the fan acoustically and mechanically with a damping chamber and flexible hosing. A control valve was used to control the flow rate.

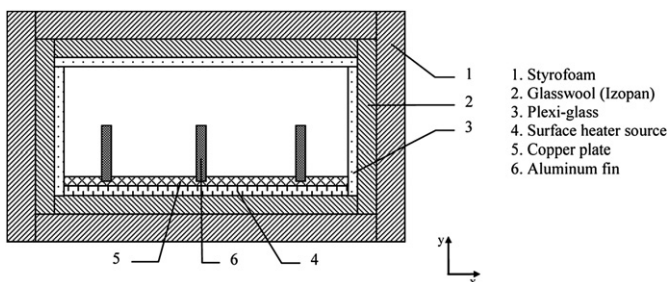


Fig. 7. Cross section view of the channel.

A digital multimeter with accuracy of $\pm 1\%$ for voltage and $\pm 1\%$ for resistance was utilized. Sixty four copper–constantan thermocouples were used in total. All thermocouples were separately calibrated. The test section was instrumented with 44 thermocouples located on the copper plate. The thermocouples on the heated section were inserted through holes drilled in the insulation and were pushed into drilled cavities placed inside the copper plate and soldered for rigidity. Temperatures were also measured at the inlet, outlet, ambient at several locations on the top, bottom and side walls of the insulation applied. The temperature signals were transferred to two 32-channel data acquisition units, and finally, sent to a PC for further processing. Temperatures measured for 15-s intervals were collected, stored and analyzed in this PC. The air velocity was measured with a hot-wire anemometer with the accuracy of ± 0.015 m/s. It was observed that the experimental conditions reach a steady-state condition about approximately 7–8 h. After conditions had been steady for some time and differences in temperatures between two intervals become negligible ($\Delta T < 0.1$ °C), all temperatures were collected, averaged and stored.

3.2. Processing of the experimental data

The data obtained during the experiments are values of temperature, velocity, voltage drop across the heater, and electric current. Using these values, the average Nusselt number for the heated bottom plate is defined as,

$$Nu_{av} = \frac{h_{av} D_h}{k_{air}} \quad (14)$$

where, D_h and k_{air} are the average heat transfer coefficient, characteristic length, and thermal conductivity of air, respectively. The average heat transfer coefficient h_{av} and D_h are defined below.

$$h_{av} = \frac{Q_{convection}}{A_b(T_w - T_{in})} \quad (15)$$

$$D_h = \frac{4A_c}{P} \quad (16)$$

In Eq. (15), $Q_{convection}$ is the convection heat transfer rate to the fluid. $A_b = L \cdot W$ is unfinned surface area of heated plate. T_w and T_{in} are average copper plate surface temperature and inlet temperature of air, respectively. A_c is channel cross sectional area, and P is the perimeter of the channel. Convection heat transfer from both the bottom copper plate and fins ($Q_{convection}$) was determined from an energy balance,

$$Q_{total} = Q_{convection} + Q_{conduction} + Q_{radiation} \quad (17)$$

where Q_{total} is total dissipated energy from the surface heater source, $Q_{conduction}$ is the total conduction heat loss through the insulation of test section, and $Q_{radiation}$ is the total radiation heat loss from bottom plate and fin surfaces. The total dissipated energy was determined from ohm's law, $Q_{total} = VI$ at the heater source. The voltage drop V and current I were measured during the experiment. The heat loss through the insulation of test section was calculated from

$$Q_{conduction} = -k_{insulation} A_{insulation} \frac{\Delta T_{insulation}}{L_{insulation}} \quad (18)$$

where $k_{insulation}$ is the thermal conductivity of the insulation. $\Delta T_{insulation}$ is the difference between internal and external surface temperature of the insulation. Radiation losses were determined from the assumption of isothermal, gray, diffuse and opaque surfaces. These losses were determined by using the method of Ellison [19] and the following equation:

$$Q_{radiation} = F A_R \sigma (T_w^4 - T_b^4) \quad (19)$$

where F is the gray body shape factor, A_R is the surface area for radiation heat transfer, and σ Stephan–Boltzmann constant. The results of experiments showed that the radiation losses never exceeded 6% of the total power dissipated. Total heat losses, however, were calculated to be always less than 13% of the total power dissipated.

The dimensionless numbers affecting the heat transfer are,
The Reynolds number:

$$Re = \frac{w_{in} D_h}{\nu_{air}} \quad (20)$$

The modified Grashof number Gr^* :

$$Gr^* = \frac{g \beta q_{con} D_h^4}{k_{air} \nu_{air}^2} \quad (21)$$

where $q_{con} = \frac{Q_{convection}}{A_b}$ is average convection heat flux transferred to the fluid. The fluid properties used in these definitions were determined at the arithmetic average of average copper plate and fluid inlet $\left(\frac{T_w + T_{in}}{2}\right)$.

In order to determine the reliability of the experimental results an uncertainty analysis was conducted on all measure quantities as well as the quantities calculated from the measurement results. Uncertainties were estimated according to the standard procedures

Table 2
Fin array configurations.

Fin-array set	Fin height (H_f/H)	Clearance parameter $C = c/H$	Fin spacing (S/H)
1	0.25	0.75	0.04
2	0.25	0.75	0.08
3	0.25	0.75	0.12
4	0.25	0.75	0.18
5	0.50	0.50	0.04
6	0.50	0.50	0.08
7	0.50	0.50	0.12
8	0.50	0.50	0.18
9	0.80	0.20	0.04
10	0.80	0.20	0.08
11	0.80	0.20	0.12
12	0.80	0.20	0.18

reported in the literature [29,30]. Overall, the uncertainty in the Nusselt number is around $\pm 6\%$ and for the modified Grashof number it is around $\pm 4\%$, which is primarily due to uncertainties in the convective heat flux values. Uncertainty in the Reynolds number is around $\pm 2.5\%$.

4. Results and discussion

The laminar mixed convection heat transfer with longitudinal fins in a horizontal channel under the bottom wall constant heat flux conditions has been investigated numerically and experimentally. The results are presented using the modified Grashof Number Gr^* based on the heat flux and the channel geometry. Numerical results were obtained for nondimensional fin spacings ranging between $0.04 \leq S/H \leq 0.18$ and for nondimensional fin heights ranging between $0.25 \leq H_f/H \leq 0.80$. The fin array configurations studied are listed in Table 2.

The variation of average heat transfer coefficient with clearance parameter is presented in Figs. 8 to 10 for different modified Grashof numbers and Reynolds numbers for the fin spacing $S/H = 0.12$. As can be seen from these figures the average heat transfer coefficient, increases with the decrease in clearance parameter. For a fixed value of channel height and constant mass flow rate, if the clearance gap between the fin tip and channel top wall is decreased (decreasing clearance parameter), fins approaching the top surface cause significant enhancement in the heat transfer coefficient by forcing all the flow into the region formed by the fins. If the clearance gap is increased, the cold fluid flows above the inner fin region causing the hot fluid to stay between fin arrays and therefore heat transfer decreases.

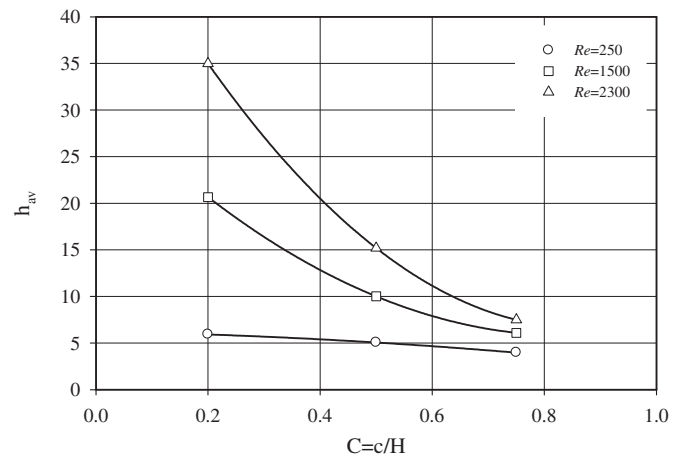


Fig. 8. Variation of heat transfer coefficient with clearance parameter for different Reynolds numbers for $S/H = 0.12$ and $Gr^* = 5 \times 10^7$.

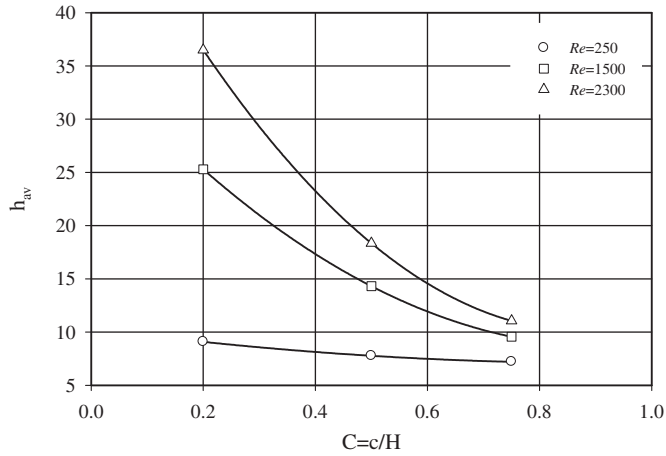


Fig. 9. Variation of heat transfer coefficient with clearance parameter for different Reynolds numbers for $S/H = 0.12$ and $Gr^* = 5 \times 10^8$.

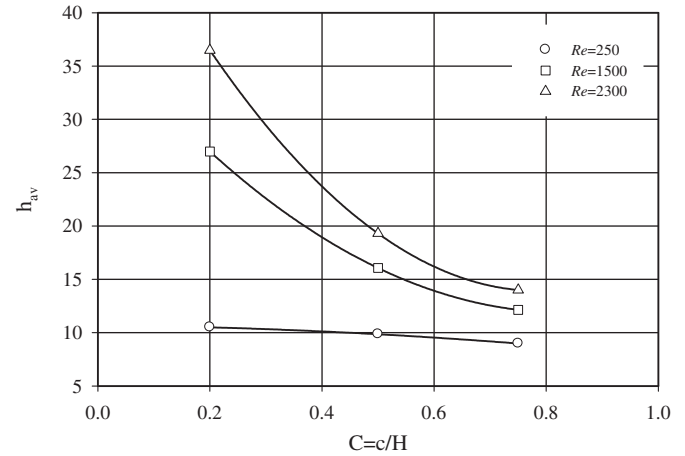


Fig. 10. Variation of heat transfer coefficient with clearance parameter for different Reynolds numbers for $S/H = 0.12$ and $Gr^* = 1 \times 10^9$.

This effect can be clearly seen from numerical results. Figs. 11 and 12 present velocity vectors and streamlines at the same Reynolds number, modified Grashof number, fin spacing and the same location ($x = 0.00325$ m) for $C = 0.75$ and $C = 0.20$, respectively. As it can be clearly seen from the velocity vectors and streamlines, a parallel flow exists in the direction of main flow until the channel formed by the fins in both cases ($C = 0.75$ and $C = 0.20$). For $C = 0.20$, the fluid continues flowing at the inner fin region in the main flow direction. On the contrary, for $C = 0.75$ the fluid tries to escape from the inner fin region to the outside region. Consequently, the cold fluid which is entering the inner fin region leaves the channel without absorbing much heat from the heated surface of the duct. This can be one of the main reasons why the heat transfer coefficient values decrease with the increase in clearance parameter. This result can also be obtained by observing Fig. 13 that shows temperature distributions for the same case but for the different clearance parameters of $C = 0.75$ and $C = 0.20$. For $C = 0.75$, near the end of the inner fin region a large part of the duct is occupied by heated fluid which prevents the heat transfer to be

achieved efficiently. However, for $C = 0.20$, the fluid removes much heat from the same region. The temperature distribution on z - y plane of Fig. 13b is also presented on x - y plane in Fig. 14. Temperature distributions in the copper plate, fin and fluid region can be seen more clearly in this figure.

Under mixed convection conditions, flow is driven by an externally imposed pressure gradient, as well as by buoyancy forces. The buoyancy forces significantly affect the heat transfer characteristics of laminar mixed convection in a fin array. For horizontal channels if heating occurs at the bottom surface, buoyancy may induce a secondary flow which is in combination with the main flow. The buoyancy driven secondary flow leads to an increase in the effects of natural convection and therefore, causes an increase in the rate of heat transfer. If the clearance gap is reduced, the buoyancy driven secondary flows can be developed. Because, the decrease in the clearance gap provides the required pressure decrease in the region bounded by fin plates, which causes the secondary flows to develop and an increase in the rate of heat transfer.

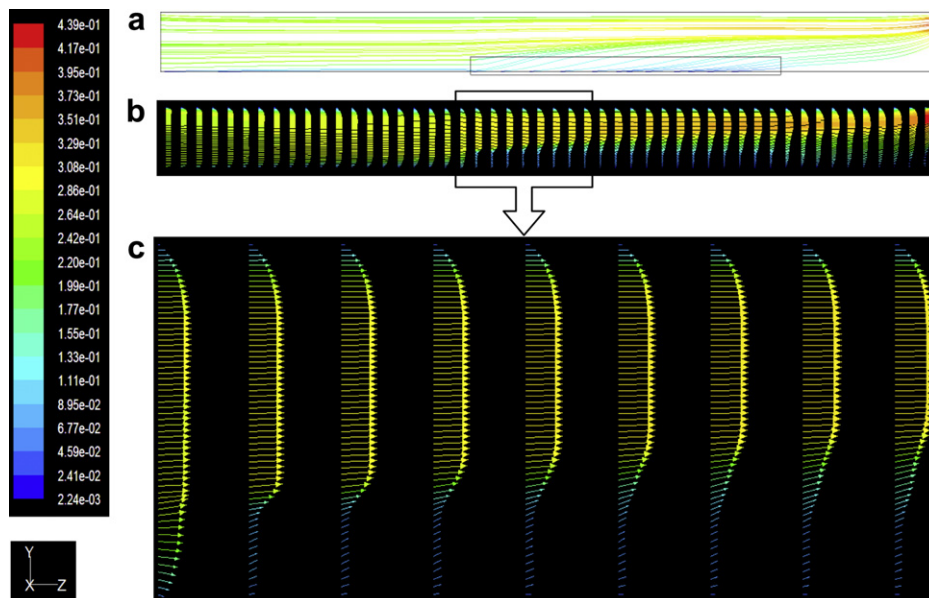


Fig. 11. Velocity vector and streamlines for $Re = 2300$, $Gr^* = 1 \times 10^9$, $S/H = 0.12$ and $C = 0.75$ at $x = 0.00325$ m in the y - z plane (a) streamlines, (b) velocity vector, (c) partially enlarged view of velocity vectors.

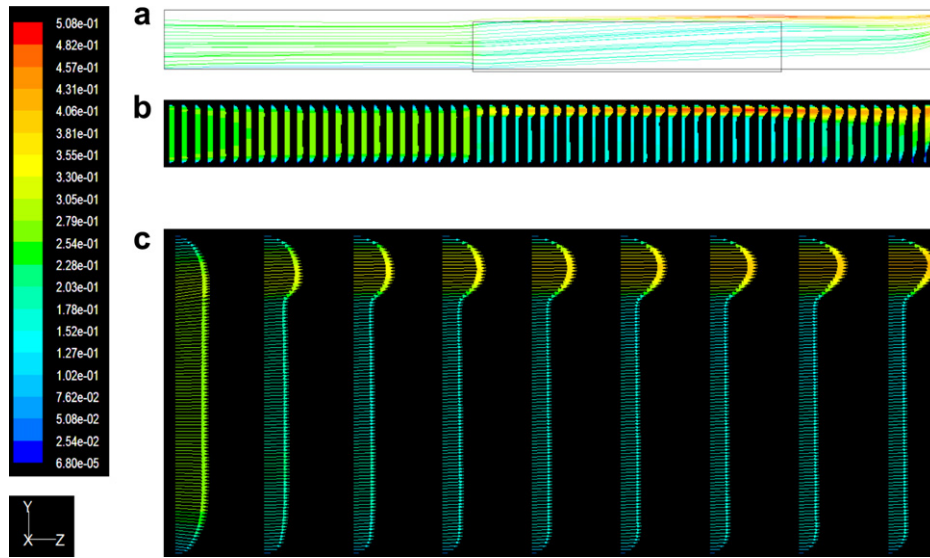


Fig. 12. Velocity vector and streamlines for $Re = 2300$, $Gr^* = 1 \times 10^9$, $S/H = 0.12$ and $C = 0.20$ at $x = 0.00325$ m in the y – z plane (a) streamlines, (b) velocity vector, (c) partially enlarged view of the velocity vectors.

At a low Reynolds number ($Re = 250$), however, as seen from Figs. 8–10 a small increase in the average heat transfer coefficient is obtained with decreasing clearance parameter. Increasing the Reynolds number (increasing fluid velocity), causes an increase in forced convection effects on heat transfer, which results in a significant increase in the average heat transfer coefficient. From examination of these figures, it can also be seen that, at high values of the clearance parameter, the increase in the average heat transfer coefficient with Reynolds number is insignificant compared to the case at low values of the clearance parameter. When the clearance parameter is decreased from $c/H = 0.75$ to $c/H = 0.50$, for the same mass flow rate at every fin spacing, for high values of Reynolds number, the percentage increase in average heat transfer coefficient is about 50%. However, as the clearance parameter is decreased from $c/H = 0.50$ to $c/H = 0.20$, the percentage increase is about 100%. The reason of this, as explained above, if the clearance gap is decreased the buoyancy driven secondary flows develop and consequently, the average heat transfer coefficient increases. Furthermore, the decrease in the clearance gap causes much of the fluid to flow through the channels formed by the fins, which results in a significant increase in the average heat transfer coefficient.

Figs. 15–17 present variations of the average convection heat transfer coefficient with fin spacing for different modified Grashof numbers and clearance parameters, $C = 0.20$, 0.50 and 0.75 . These figures show that, the increase or decrease in fin spacing after

a fixed value does not cause an enhancement in the heat transfer coefficient. On the contrary, it causes a reduction in a heat transfer coefficient. The average heat transfer coefficient first increases with fin spacing up to a maximum value and then it decreases with the increase in fin spacing. The value of fin spacing at which the heat transfer takes its maximum value, is defined as optimum fin spacing, S_{opt} . These figures show that the optimum fin spacing is between 8 and 12 mm.

As can be seen from these figures, when the heat transfer surface area is increased (increasing number of fins or decreasing fin spacing), the heat transfer coefficient decreases. The decrease in fin spacing causes an intersection between boundary layers developed on fin surfaces. The intersection of boundary layers prevents the cold fluid to enter fin grooves and thus the hot fluid stays much longer between fin arrays. Therefore, a large part of the duct is occupied by the heated fluid which prevents the heat transfer to be achieved efficiently. After the optimum value of spacing, increasing the fin spacing (decreasing the number of fins), causes a decrease in total heat transfer surface area, which results in a decrease in the rate of heat transfer. As seen from these figures in the design of fin arrays on a horizontal surface, to achieve the maximum amount of heat transfer, the fin spacing must have an optimum value. There is optimum fin spacing for each clearance parameter. At a high value of clearance parameter, the optimum fin spacing is not clearly visible. However, at a low value of clearance parameter, the optimum fin spacing is clearly visible.

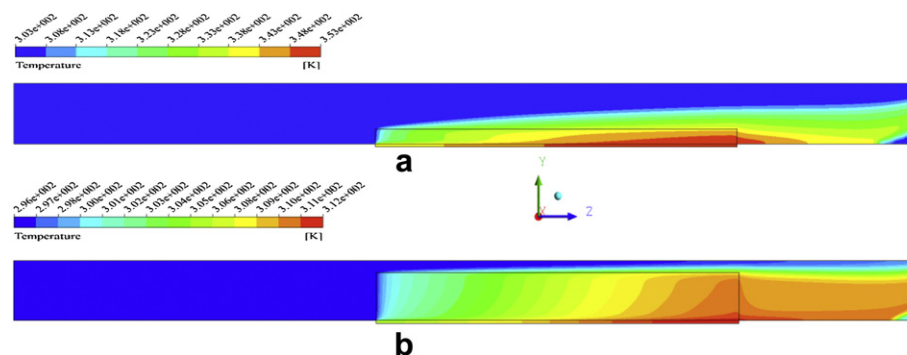


Fig. 13. Temperature distribution for $Re = 2300$, $Gr^* = 1 \times 10^9$, $S/H = 0.12$ at $x = 0.00325$ m in the y – z plane (a) $C = 0.75$ (b) $C = 0.20$.

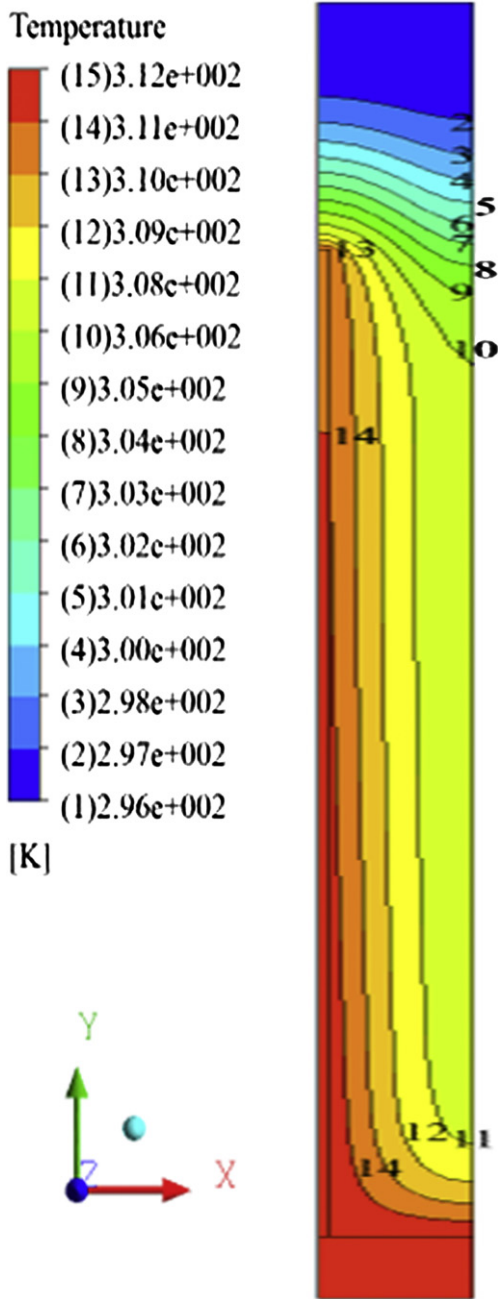


Fig. 14. Temperature distribution and contour for $Re = 2300$, $Gr^* = 1 \times 10^9$, $S/H = 0.12$ and $C = 0.20$ at $z = 1.0$ m in the x - y plane.

The variation of the average heat transfer coefficients with clearance parameter is presented in Figs. 18–20 for different modified Grashof numbers and for dimensionless fin spacings of $S/H = 0.04, 0.08, 0.12$ and 0.18 . The average heat transfer coefficient increases with the decrease in clearance parameter for each fin spacing. Increasing the fin height for a fix value of channel height causes an increase in the heat transfer surface area. Consequently, the surface temperatures of fins and copper plate decrease, which results in an increase in the heat transfer coefficient. In addition, increasing the fin height causes the buoyancy driven secondary flows to be developed greatly for the cases at which natural convection effects are important. This results in a further increase in the heat transfer coefficient.

As discussed above, if the fin height is decreased, fluid flows above the inner fin region causing the hot fluid to stay between fin

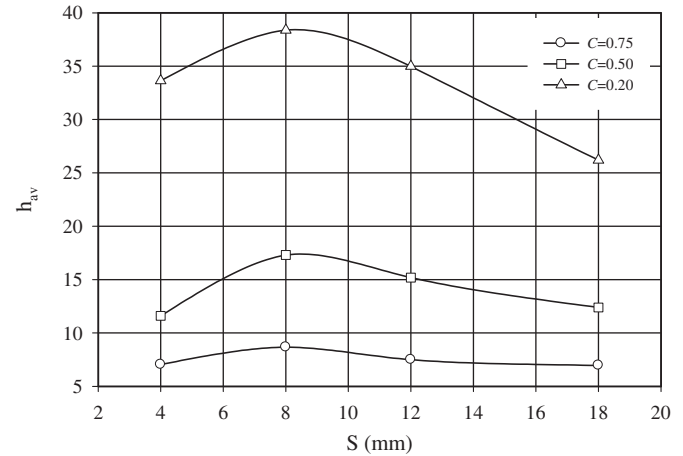


Fig. 15. Variation of heat transfer coefficient with fin spacing for different clearance parameters for $Re = 2300$ and $Gr^* = 5 \times 10^7$.

arrays and therefore, heat transfer decreases. This effect can be seen in Figs. 18–20. In addition, for clearance parameters $c/H = 0.20$ and $c/H = 0.50$ the value of the average heat transfer coefficient is greater for the fin spacing $S/H = 0.18$ than for the fin spacing $S/H = 0.04$. However, the average heat transfer coefficient obtained for the fin spacing of $S/H = 0.04$ is greater for the clearance parameter $c/H = 0.75$. As described above in detail, this is because the buoyancy driven forces become strong enough with the increase in the fin height, and therefore, buoyancy driven secondary flows have more important effects on the heat transfer coefficient.

For the same modified Grashof number, fin spacing and height of the channel, forced convection effects on the heat transfer increase with the decrease in clearance parameter. As a result of the decrease in clearance parameter, a large amount of fluid flow goes through the inner fin region, and the heat transfer coefficient increases. This effect can clearly be seen from Figs. 18–20. From examination of these figures, it can also be seen that at high values of the clearance parameter, the effect of fin spacing is insignificant compared to the cases at low values of the clearance parameter.

It has been shown in the present work that the average Nusselt number depends on the Reynolds number, modified Grashof number, fin spacing, fin height, tip clearance and fin length. Basing

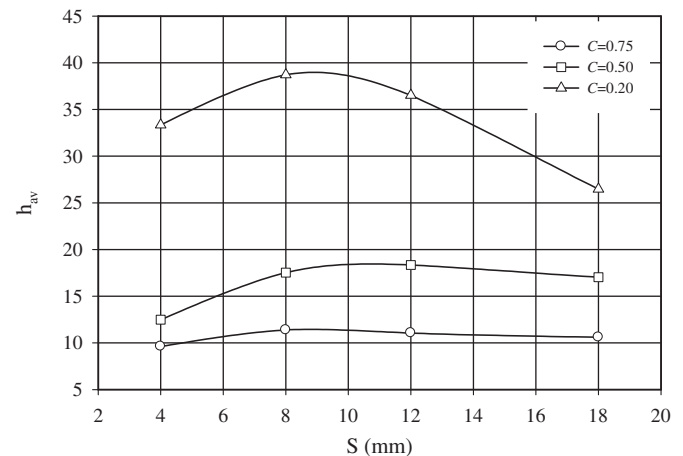


Fig. 16. Variation of heat transfer coefficient with fin spacing for different clearance parameters for $Re = 2300$ and $Gr^* = 5 \times 10^8$.

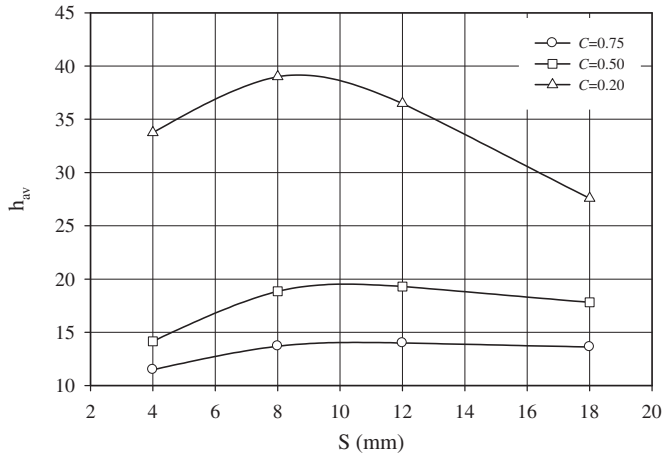


Fig. 17. Variation of heat transfer coefficient with fin spacing for different clearance parameters for $Re = 2300$ and $Gr^* = 1 \times 10^9$.

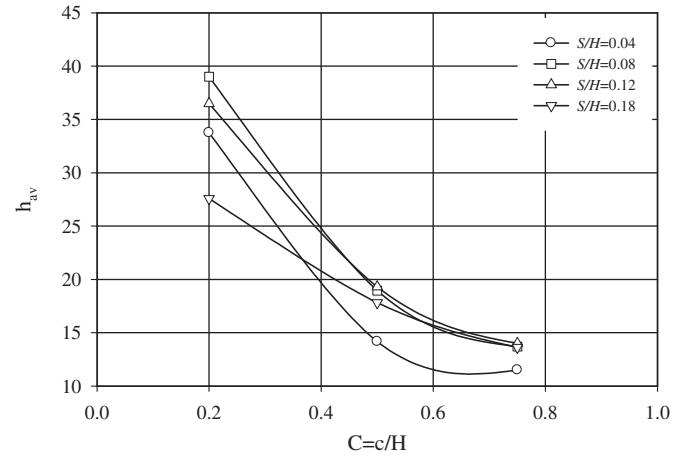


Fig. 20. Variation of heat transfer coefficient with clearance parameter for different dimensionless fin spacings for $Re = 2300$ and $Gr^* = 1 \times 10^9$.

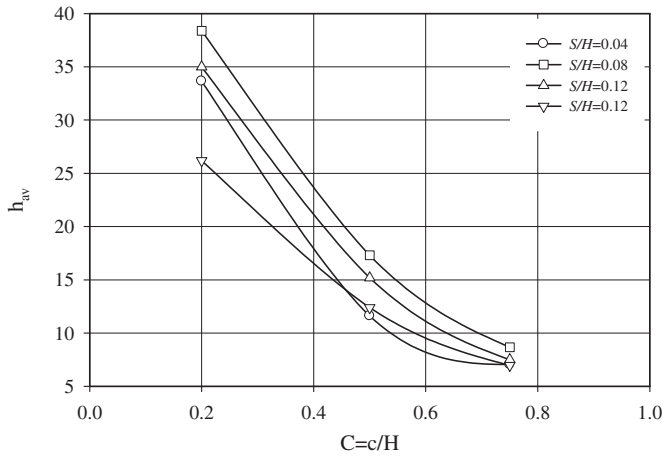


Fig. 18. Variation of heat transfer coefficient with clearance parameter for different dimensionless fin spacings for $Re = 2300$ and $Gr^* = 5 \times 10^7$.

on the Levenberg–Marquardt algorithm, and considering the effects of above parameters, an empirical equation from experimental results has been derived to correlate the average Nusselt number. The correlation could be expressed by the equation

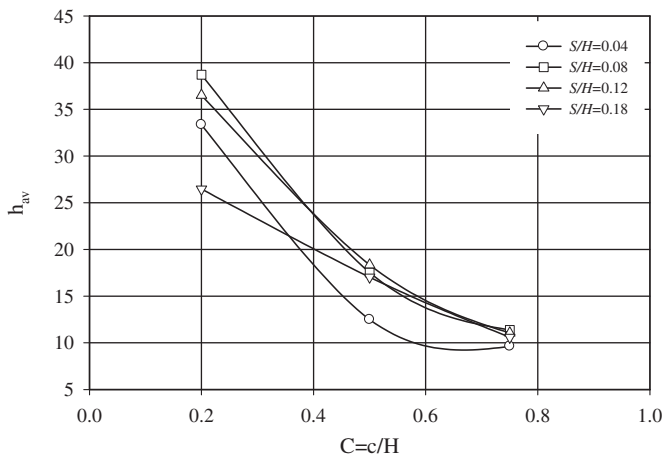


Fig. 19. Variation of heat transfer coefficient with clearance parameter for different dimensionless fin spacings for $Re = 2300$ and $Gr^* = 5 \times 10^8$.

$$Nu_{av} = 0.06312(Re)^{0.5639}(Gr^*)^{0.0843}\left(\frac{S}{H_f}\right)^{-0.0312} \times \left(\frac{c}{H_f}\right)^{-0.621}\left(\frac{L_f}{H_f}\right)^{0.50} \quad (22)$$

within the ranges $250 \leq Re \leq 2300$, $5 \times 10^7 \leq Gr^* \leq 1 \times 10^9$, $0.05 \leq S/H_f \leq 0.72$, $0.25 \leq c/H_f \leq 3.0$, and $7.5 \leq L_f/H_f \leq 24.0$. The relative error for this correlation is $\pm 18\%$. The reader should note that this correlation is only valid for thin fins, which do not cause much effect on longitudinal air flow. And this correlation does not show effects of fin cross sectional area, thermal conductivity and heat transfer coefficient of the fin.

An equation has also been developed from the numerical data obtained to express the average Nusselt number in the terms of Re , Gr^* , (S/H_f) , (c/H_f) and (L_f/H_f) . This equation developed by simple regression is:

$$Nu_{av} = 0.0603(Re)^{0.5639}(Gr^*)^{0.0843}\left(\frac{S}{H_f}\right)^{-0.0312} \times \left(\frac{c}{H_f}\right)^{-0.621}\left(\frac{L_f}{H_f}\right)^{0.50} + 7.637 \quad (23)$$

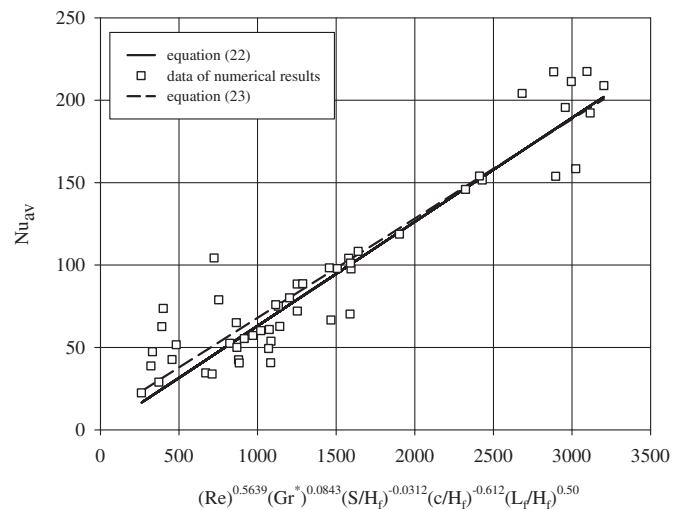


Fig. 21. Comparison of the correlation obtained from the experimental results with numerical results.

A comparison of numerical results and the equation (23), with the correlation obtained from experimental results in the present study is shown in Fig. 21. This figure shows that, there is a good agreement between the equation (23) and the correlation equation.

5. Conclusions

The influence of the clearance gap between fin tips and top surface of channel on heat transfer from longitudinal fins inside a horizontal channel has been experimentally and numerically investigated for a wide range of modified Grashof numbers and different clearance parameters and spacings. A conjugate analysis has been carried out in which the conservation of mass, momentum and energy equations for the fluid in the two fin enclosure are solved together with the heat conduction equation in the fin and the base plate. The results for bottom heated fin arrays have been presented for different fin spacings, clearance parameters, and modified Grashof numbers, and the effects of these parameters on heat transfer have been investigated.

It has been determined that the mixed convection heat transfer depends on the fin height, clearance gap and spacing. The effects of fin spacing on heat transfer have been investigated numerically at four different dimensionless fin spacings ($S/H = 0.04, 0.08, 0.12$ and 0.18). The average convection heat transfer coefficient increases first with fin spacing and then it takes its maximum value after which it starts to decrease with the increase in fin spacing. When the fin spacing is smaller than the required value, the resistance against the flow is formed due to the intersection of boundary layers developed on fin surfaces and as a result, the rate heat transfer from fin arrays decreases. For large values of fin spacing (a small number of fins for a fixed fin base area), however, the decrease in the total heat transfer area causes the rate of heat transfer to decrease. The results of this study have shown that to obtain the maximum amount of heat transfer from fin arrays, the fin spacing should be at an optimum value. The optimum fin spacing has been obtained in this study as $S_{opt} = 8\text{--}12$ mm.

A correlation for average Nusselt number has also been presented to relate the heat transfer from fin arrays in the channel with dimensionless, Reynolds number, modified Grashof number and other experimental parameters such as fin spacing, fin height, tip clearance and fin length.

Acknowledgements

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References

- [1] F.P. Incropera, Convection heat transfer in electronic equipment cooling, *Journal of Heat Transfer* 110 (1998) 1097–1111.
- [2] W. Elenbaas, Heat dissipation of parallel plates by free convection, *Physica* 9 (1942) 1–28.
- [3] K.E. Starner, H.N. McManus, An experimental investigation of free convection heat transfer from rectangular fin-arrays, *Journal of Heat Transfer, Transactions of the ASME Series C* 85 (1963) 273–278.
- [4] J.R. Welling, C.V. Wooldridge, Free convection heat transfer coefficients from rectangular fin arrays, *ASME Journal of Heat Transfer* 87 (1965) 439–444.
- [5] F. Harahan, H.N. McManus, Natural convection heat transfer from rectangular fin arrays, *Journal of Heat Transfer, Transactions of the ASME Series C* 89 (1967) 32–38.
- [6] Charles D. Jones, Lester F. Smith, Optimum arrangement of rectangular fins on horizontal surfaces for free convection heat transfer, *Journal of Heat Transfer, Transactions of the ASME Series C* 92 (1970) 6–10.
- [7] H. Yuncu, G. Anbar, An experimental investigation on performance of rectangular fins on a horizontal base in free convection heat transfer, *Heat and Mass Transfer* 33 (1998) 507–514.
- [8] B. Yazicioglu, H. Yuncu, A correlation for optimum fin spacing of vertically-based rectangular fins arrays subjected to natural convection heat transfer, *Journal of Thermal Science and Technology* 29 (1) (2009) 99–105.
- [9] H. Yuncu, O. Ekin, The optimum spacing between parallel heat generating boards cooled by laminar forced convection, *Journal of Thermal Science and Technology* 26 (2) (2006) 1–10.
- [10] A. Guvenc, H. Yuncu, An experimental investigation on performance of fins on a horizontal base in free convection heat transfer, *Heat and Mass Transfer* 37 (2001) 409–416.
- [11] B. Yazicioglu, H. Yuncu, Optimum fin spacing of rectangular fins on a vertical base in free convection heat transfer, *Heat and Mass Transfer* 44 (2007) 11–21.
- [12] H.G. Yalcin, S. Baskaya, M. Sivrioglu, Numerical analysis of natural convection heat transfer from rectangular shrouded fin arrays on a horizontal surface, *International Communications in Heat and Mass Transfer* 35 (2008) 299–311.
- [13] M. Mobedi, H. Yuncu, A three dimensional numerical study on natural convection heat transfer from short horizontal rectangular fin array, *Heat and Mass Transfer* 39 (2003) 267–275.
- [14] C.W. Leung, S.D. Probert, M.J. Shilston, Heat exchanger: optimal separation for vertical rectangular fins protruding from a vertical rectangular base, *Applied Energy* 19 (1985) 77–85.
- [15] C.W. Leung, S.D. Probert, M.J. Shilston, Heat-exchanger design: optimal separation between rectangular fins Protruding from a vertical rectangular base, *Applied Energy* 19 (1985) 287–299.
- [16] C.W. Leung, S.D. Probert, M.J. Shilston, Heat-exchanger design: thermal performances of rectangular fins protruding from a vertical or horizontal rectangular bases, *Applied Energy* 20 (1985) 123–140.
- [17] Senol Baskaya, Mecit Sivrioglu, Murat Ozek, Numerical flow visualization of natural convection from horizontal rectangular fin arrays, *Journal of Thermal Science and Technology* 25 (2) (2005) 22–28.
- [18] Senol Baskaya, Mecit Sivrioglu, Murat Ozek, Parametric study of natural convection heat transfer from horizontal rectangular fin arrays, *International Journal of Thermal Sciences* 39 (2000) 797–805.
- [19] Gordon N. Ellison, Generalized computations of the gray body shape factor for thermal radiation from a rectangular U-channel, *IEEE Transactions on Components Hybrids and Manufacturing Technology* CHMT 2 (4) (1979) 517–522.
- [20] S. Acharya, S.V. Patankar, Laminar mixed convection in a shrouded fin array, *ASME Journal of Heat Transfer* 103 (1981) 559–565.
- [21] I.M. Rustum, H.M. Soliman, Experimental investigation of laminar mixed convection in tubes with longitudinal internal fins, *ASME Journal of Heat Transfer* 110 (1988) 366–372.
- [22] Saad A. El-Sayed, Shamloul M. Mohamed, Ahmed M. Abdel-latif, Abdel-hamid E. Abouda, Investigation of turbulent heat transfer and fluid flow in longitudinal rectangular-fin arrays of different geometries and shrouded fin array, *Experimental Thermal and Fluid Science* 26 (2002) 879–900.
- [23] E.A.M. Elshafei, Effect of flow bypass on the performance of a shrouded longitudinal fin array, *Applied Thermal Engineering* 27 (2007) 2233–2242.
- [24] D.S. Kadle, E.M. Sparrow, Numerical and experimental study of turbulent heat transfer and fluid flow in longitudinal fin array, *ASME Journal of Heat Transfer* 108 (1986) 16–23.
- [25] E.M. Sparrow, D.S. Kadle, Effect of tip to shroud clearance on turbulent heat transfer from a shrouded, longitudinal fin array, *ASME Journal of Heat Transfer* 108 (1986) 519–524.
- [26] E.M. Sparrow, B.R. Baliga, S.V. Patankar, Forced convection heat transfer from a shrouded fin array with and without tip clearance, *ASME Journal of Heat Transfer* 100 (1978) 572–579.
- [27] J.R. Maughan, F.P. Incropera, Mixed convection heat transfer with longitudinal fins in a horizontal parallel plate channel: part II – experimental results, *ASME Journal of Heat Transfer* 112 (1990) 619–624.
- [28] M. Dogan, Numerical and Experimental investigation of heat transfer with longitudinal fins in a channel under laminar mixed convection conditions, PhD thesis, Gazi University, Ankara, Turkey, 2007.
- [29] R.J. Moffat, Contributions to the theory of single-sample uncertainty analysis, *Journal of Fluids Engineering* 104 (1982) 250–260.
- [30] R.J. Moffat, Using uncertainty analysis in the planning of an experiment, *Journal of Fluids Engineering* 107 (1985) 173–178.